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Solution NMR structure of Se0862, a highly conserved cyanobacterial protein involved in biofilm formation

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Abstract

Biofilms are accumulations of microorganisms embedded in extracellular matrices that protect against external factors and stressful environments. Cyanobacterial biofilms are ubiquitous and have potential for treatment of wastewater and sustainable production of biofuels. But the underlying mechanisms regulating cyanobacterial biofilm formation are unclear. Here, we report the solution NMR structure of a protein, Se0862, conserved across diverse cyanobacterial species and involved in regulation of biofilm formation in the cyanobacterium *Synechococcus elongatus* PCC 7942. Se0862 is a class α+β protein with ααββββαα topology and roll architecture, consisting of a four-stranded β-sheet that is flanked by four α-helices on one side. Conserved surface residues constitute a hydrophobic pocket and charged regions that are likely also present in Se0862 orthologs.

KEYWORDS

biofilm, cyanobacteria, NMR spectroscopy, protein structure, *S. elongatus* PCC 7942

1 | INTRODUCTION

Cyanobacteria, also referred to as blue-green algae, are ancient photosynthetic organisms and the only prokaryotes known to carry out oxygenic photosynthesis, thrive in a wide range of ecological habitats, and can produce biofilms under certain conditions. Biofilms are communities of microorganisms that are enclosed in extracellular polymeric matrices that provide protection against toxic compounds and stressful environmental factors such as extreme temperatures and pH.[1] Biofilms from cyanobacteria have been used for wastewater treatment, research of sustainable biofuels and biological by-products, and soil fertility improvement.[2–12] In spite of these advantages, they can also be responsible for biodeterioration and biofouling that cause economic loss.[13–15]

Recently, there have been advances in understanding biofilm formation by cyanobacteria. For example, four small secreted proteins, EbfG1-4 (enable biofilm formation with GG-motif), of *S. elongatus* PCC 7942 have been found to promote biofilm-formation.[16,17] Also, *Synechocystis* sp. PCC 6803 mutants lacking type IV pili

(T4P) are unable to aggregate and form biofilms,[18] however, impairment of T4P formation in *S. elongatus* resulted in biofilm-forming-mutants, unlike the wild-type strain that grow planktonically.[19] These data suggest that *S. elongatus* and *Synechocystis* employ different mechanisms for biofilm development. T4P biogenesis is under the control of the circadian clock in cyanobacteria,[20] used for extension, adhesion, retraction and motility, and plays a role in natural transformation by pulling exogenous DNA close to the cell surface.[21,22]

Synpcc7942_0862 encodes a conserved hypothetical protein, Se0862, in *S. elongatus*, and its inactivation leads to biofilm development.[19] In order to gain insights into Se0862 as a biofilm regulator we cloned the gene, expressed it in *E. coli*, purified the protein, and determined the solution structure by NMR.

2 | RESULTS AND DISCUSSION

2.1 | NMR solution structure of Se0862

Purified Se0862 elutes from a preparative size-exclusion column as a monomer (Figure S1). A two-dimensional ^{1}H-^{15}N HSQC NMR spectrum of Se0862 obtained at a static magnetic field strength of 14.1 T is shown in Figure 1a. Standard heteronuclear triple resonance NMR experiments were used to assign backbone and sidechain

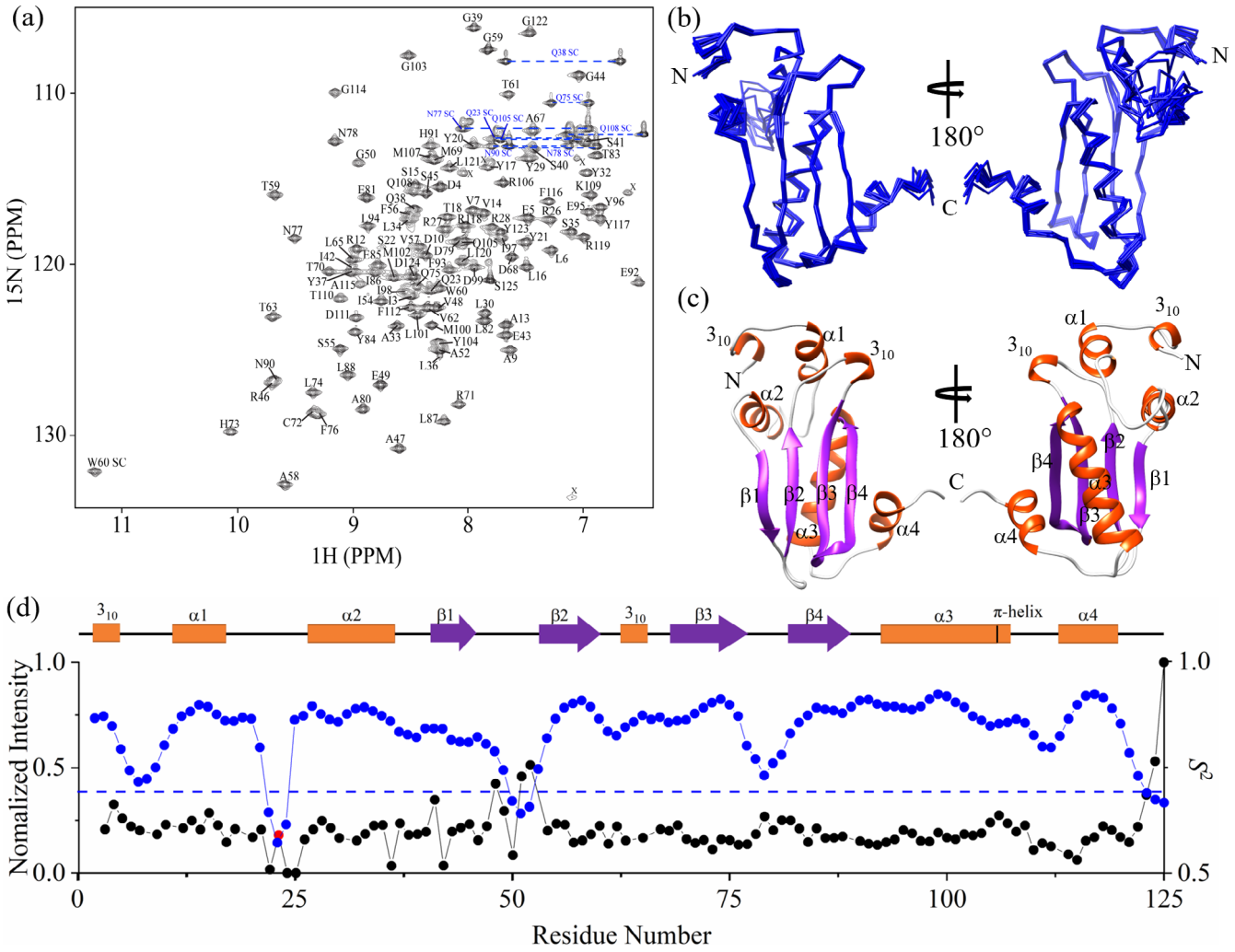


FIGURE 1 2D HSQC NMR spectrum and NMR solution structure of Se0862. (a) Assigned two dimensional ^{1}H-^{15}N HSQC spectrum of Se0862. SC, side chain. Blue dashed lines connect pairs of peaks of NH_2 groups of Asn and Gln side chains. Unassigned peaks are marked with an "x". (b) Superposition of backbone atoms of a family of 10 lowest-energy structures of Se0862. (c) Ribbon diagram of the energy-minimized average structure. (d) Plot of normalized ^{15}N,^{1}H HSQC peak intensities (black circles) and TALOS-calculated S^2 values (blue circles) as a function of residue number. Blue dash line represents S^2 value of 0.7. The backbone resonance of residue Q23 overlaps with that of W60 and is thus its intensity is denoted with a red circle. Resonances for A24 and N25 were unobservable and are thus assumed to have near-zero peak intensities

TABLE 1 Structural statistics for Se0862

Protein data Bank accession number	6UF2
BMRB accession number	30,676
Total residue number	125
NMR constraints	
Distance constraints	
Total	2045
Intra-residue	1,080
Inter-residue	
Sequential ($\|i–j\| = 1$)	553
Medium-range ($1 < \|i–j\| \leq 4$)	184
Long-range ($\|i–j\| > 4$)	228
Total dihedral angle restraints	192
Phi (ϕ)	96
Psi (ψ)	96
Total RDCs	175
HN-N	63
HA-CA	56
CA-CO	56
Q[a]	0.173
Structure statistics	
Violations (mean ± s.d.)	
Distance constraints (Å)	0.068 ± 0.056
Dihedral angle constraints (°)	0.893 ± 0.778
Dihedral angle violation (>5.0°)	0
Distance violation (>0.5 Å)	0
Deviation from idealized geometry	
Bond lengths (Å)	0.007 ± 0.000
Bond angle (°)	0.862 ± 0.016
Impropers (°)	0.573 ± 0.021
Average pairwise r.m.s.d. (Å)[b]	
Heavy atoms (all) (α-helices & β-strands)	0.85
Backbone atoms (all) (α-helices & β-strands)	0.25
Ramachandran plot (%)	
Richardson Lab's Molprobity	
Most favored regions	96.8
Allowed regions	3.2
Disallowed regions	0.0
PROCHECK	
Most favored regions	91.1
Additional allowed regions	7.4
Generously allowed regions	1.5
Disallowed regions	0.0

chemical shifts of 121 out of 125 residues. The four residues without any chemical shift assignments are M1, R2, A24, and N25. All nine prolyl residues—P8, P11, P19, P31, P53, P64, P66, P89, P113—adopt the *trans* conformation (Table S1), as based on differences between their $^{13}C^{\beta}$ and $^{13}C^{\gamma}$ chemical shifts[23] (range: 2.54–5.62 ppm; measured using the CC(CO)NH experiment[24]) and the program Promega.[25] The structure of Se0862 was calculated using XPLOR-NIH[26] with 2045 distance restraints, 192 ϕ, ψ backbone dihedral angle restraints determined by TALOS-N,[27] and 63 ^{15}N–H, 56 $^{13}C^{\alpha}$–H^{α}, and 56 $^{13}C^{\alpha}$–$^{13}C'$ residual dipolar coupling restraints. Figure 1b shows the family of the 10 lowest-energy structures out of 100. Statistics on the family of structures are listed in Table 1. There were no violations of NOE-derived distance restraints (threshold of 0.5 Å) or TALOS-derived dihedral angle restraints (threshold of 5°). The family of 10 structures of Se0862 is well-defined with a root-mean-square deviation (RMSD) of 0.25 Å for backbone atoms and 0.85 Å for all heavy atoms. The ribbon representation of the average minimized structure is shown in Figure 1c. Se0862 adopts an αααββββαα topology with two antiparallel N-terminal α-helices (α1: P11-Y17; α2: Y29-Y37), a four-stranded antiparallel β-sheet (β1: I42-K46; β2: I54-T59; β3: M69-Q75; β4: D79-P89), two antiparallel C-terminal α-helices (α3: N90-R106; α4: G114-L121), two 3_{10}-helices (I3-E5 and L65-A67) and a π-helix (Q105-R106)[28] (Figure 1c). According to CATH,[29] which is a database that classifies distinct globular domains into homologous superfamilies, Se0862 is a class α + β protein with roll architecture. Normalized ^{1}H-^{15}N HSQC peak intensities and order parameters, S^2, derived from chemical shifts[27,30] are plotted in Figure 1d and suggest that two type I turns between β1 and β2 and between β3 and β4 experience local dynamics on the fast time scale, whereas the loop between α1 and α2 also undergoes motions on the intermediate time scale.

2.2 | Structural properties and potential function of Se0862

The α-helices within each antiparallel pair pack together through mostly hydrophobic interactions, as do the

[a]Quality factor (or *Q* factor) is used to quantify the extent of agreement between a calculated structure and measured dipolar couplings. A *Q* factor below 0.2 can be used as a rule of thumb to indicate good agreement.

[b]For RMSD calculation, residues 13–18, 28–59, 69–107, 114–120 of Se0862 were used.

β-strands within the sheet. The helices pack against one face of the β-sheet through interactions between nonpolar side chains. The hydrophobicity and solvent accessible surface area (SASA) of residues were calculated using ProtScale[31] and GETAREA[32] tools, respectively, and show that almost all conserved hydrophobic residues are buried whereas charged residues tend to be solvent exposed (Figure 2a), as is to be expected.

Several nonpolar residues, including I42, F56, M100, and L101, form a hydrophobic pocket (Figure 2b), and because many of these residues are conserved (Figure S2), the structure of the pocket might be conserved as well. Also, there is a putative salt bridge between conserved residues E85 (β4) and R71 (β3) (Figure 2c). The electrostatic surface potential of Se0862, as calculated using the APBS tool in Chimera,[33] is shown in Figure 2b with conserved acidic and basic residues labeled. A negatively charged patch is formed by the following conserved residues: residues D68, E92, E95, and D99. Two positively charged patches—(a) R27, R28, and R46, and (b) R118 and R119—are also composed of conserved residues and are separated by the negatively charged patch. The total solvent accessible surface area is 8,640 Å^2 according to PDBePISA.[34]

Using Se0862 as query, PSI-BLAST identified 478 proteins with sequence identities over 40%, all from cyanobacteria, showing that Se0862 orthologs are widespread in these organisms. However, they are designated as hypothetical proteins and missing structural and functional information. A DALI[35,36] search of the Protein Data Bank identified 123 polypeptide chains with Z-scores above 2 but lower than 3.4 (Z-score > 2 suggests that

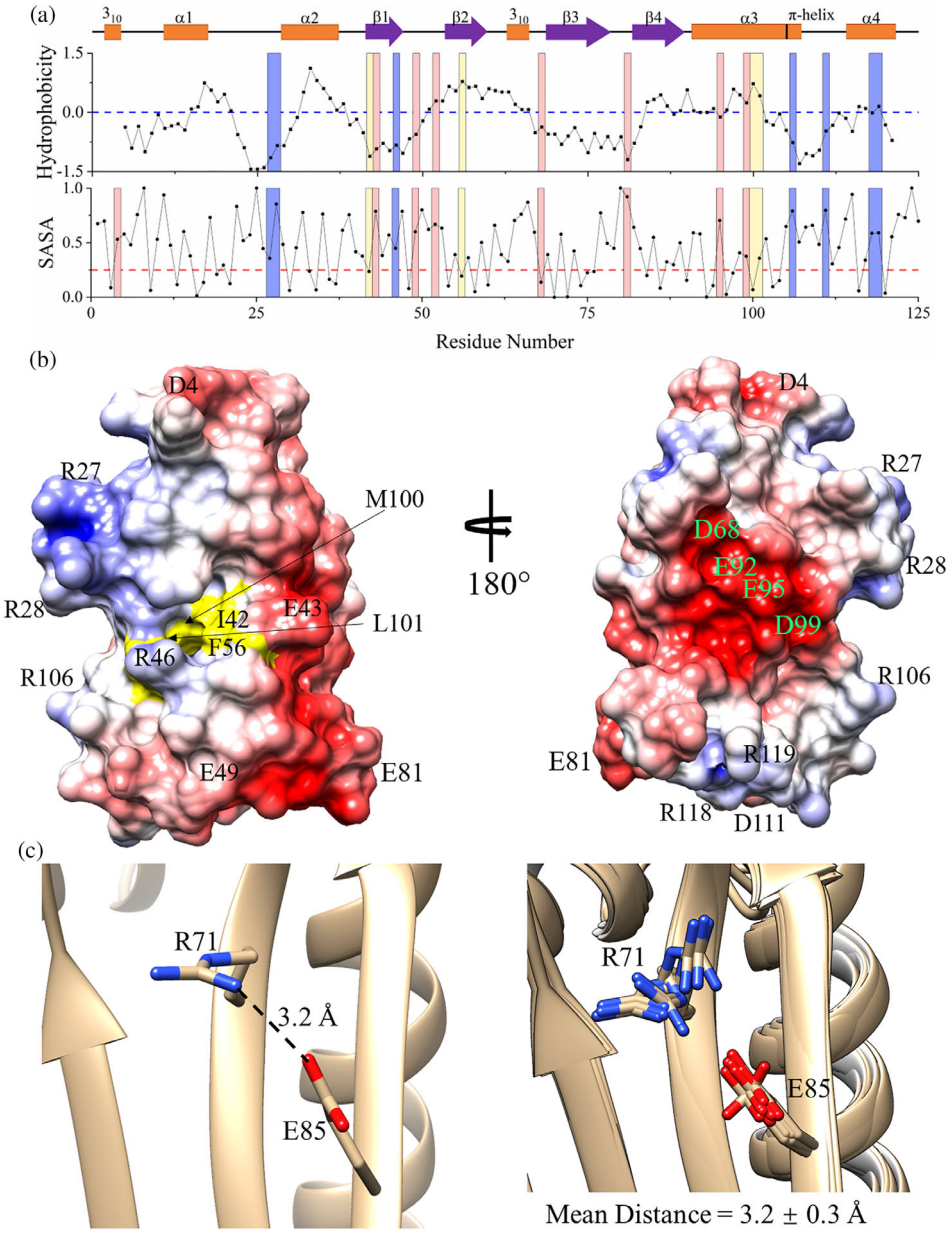


FIGURE 2 Surface properties of Se0862. (a) Hydrophobicity and SASA plots of Se0862 as a function of its primary sequence. Light yellow, red, and blue rectangles indicate conserved hydrophobic, acidic, and basic residues, respectively. Dashed horizontal lines represent 0 hydrophobicity and 0.25 SASA. (b) Hydrophobic pocket (rendered yellow) and electrostatic surface potential of Se0862 with conserved residues labeled. The scale for the surface potential is from $-5\ k_BT/e$ (red) to $+5\ k_BT/e$ (blue). (c) Side chains of E85 and R71 in the energy-minimized average structure (left) and 10 lowest-energy structures (right). The indicated distance is between an O^{ε} atom of E85 and an N^{η} atom of R71

structural similarity is significant). The top 20 DALI hits are listed in Table S2 but they did not provide clear insights into the likely function of Se0862.

Inactivation of Synpcc7942_0862 causes biofilm formation[19] and the protein encoded by this gene is also required for natural competence, the latter of which is known to need T4P.[20] A testable hypothesis is that the negatively and positively charged regions and hydrophobic pocket of Se0862 are sites of interaction with complementary sites on T4P components to regulate biofilm formation, pili biogenesis, and/or extension in cyanobacteria.

3 | MATERIALS AND METHODS

3.1 | NMR spectroscopy

All NMR experiments were carried out at a calibrated sample temperature of 22°C on a Bruker Avance III 600 MHz spectrometer equipped with a TCI cryoprobe. ^{1}H chemical shifts were referenced to internal 3-(trimethyl-silyl)-1-propanesulfonic acid sodium salt (DSS). ^{13}C and ^{15}N chemical shifts were indirectly referenced to DSS using absolute frequency ratios listed on the BMRB website. NMR samples contained ~0.6 mM ^{15}N/^{13}C-enriched protein in 350 μL NMR buffer (20 mM Tris, 100 mM NaCl, 5 mM TCEP, 0.02% NaN_3, 10 μM DSS, pH 7.0, 95% H_2O/5% D2O). 3D HNCACB, HN(CO)CACB, HNCO, HN(CA)CO, HBHA(CO)NH, HCCH(CO)NH, CC(CO)NH, (H)CCH-COSY, H(C)CH-COSY, (H)CCH-TOCSY experiments were used for backbone and aliphatic side-chain assignments[37]; 2D (HB)CB(CDCG)HG and (HB)CB(CDCGCE)HE experiments were acquired for aromatic side chain assignments.[38] 3D ^{15}N-edited, ^{13}C-edited NOESY-HSQC, and 4D ^{13}C /^{13}C-edited NOESY-HSQC spectra were collected for interproton distance restraints for structure determination. A chemical shift-based structural model of Se0862 generated by CS-ROSETTA was used to guide initial NOE assignments. Please see Table S3 for further details on NMR experiments. NMR data were processed with NMRPipe[39] and analyzed using NMRFAM-Sparky.[40] All details on cloning, expression and purification of Se0862 are presenting in supplementary materials.

3.2 | NMR structure calculations

Backbone dihedral angle restraints were predicted from backbone chemical shifts using TALOS-N.[27] All NOE distance restraints were grouped into four ranges (strong, medium, weak, very weak) according to NOE cross-peak intensities. The lower distance bound for all distance restraints was set to 1.8 Å. The upper bound was 2.7 Å for strong NOEs, 3.3 Å for medium NOEs, 5.0 Å for weak NOEs, and 6.0 Å for very weak NOEs. For methyl group NOEs an additional 0.5 Å was added to the upper bound of distance restraints. Structures were calculated with XPLOR-NIH.[26] During the high temperature stage of calculations, force constants for distances (k_{NOE}), dihedral angles (k_{CDIH}), torsion angles (k_{tDB}), implicit solvation (EEFx, k_{EEFx}), angles (k_{ANGL}), and improper torsions (k_{IMPR}) were 2 kcal mol^{-1} Å^{-2}, 10 kcal mol^{-1} Å^{-2}, 0.02 kcal mol^{-1} Å^{-2}, 0.1 kcal mol^{-1} Å^{-2}, 0.4 kcal mol^{-1} Å^{-2}, and 0.1 kcal mol^{-1} Å^{-2}, respectively. In the simulated annealing stage, k_{NOE}, k_{tDB}, k_{EEFx}, k_{ANGL}, and k_{IMPR} were gradually increased from 2 to 30 kcal mol^{-1} Å^{-2}, 0.1 to 1.0 kcal mol^{-1} rad^{-2}, 0.1 to 1.0 kcal mol^{-1} Å^{-2}, 0.4 to 1.0 kcal mol^{-1} rad^{-2}, and 0.1 to 1.0 kcal mol^{-1} Å^{-2}, respectively, and k_{CDIH} was set to 200 kcal mol^{-1} rad^{-2}. Temperature in the simulated annealing stage was decreased from 3,500 K to 25 K in steps of 12.5 K. Structure validation statistics were generated by the PSVS web server,[41] PROCHECK,[42] and MolProbity[43] (Table 1). Calculations of surface potential and hydrophobicity[44] and structural renderings were carried out by UCSF ChimeraX.[45] α-helices, β-strands, and 3_{10} and π-helices were identified using the "dssp report true" command in ChimeraX.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Ning Zhang: Data curation; formal analysis; investigation; methodology; writing-original draft; writing-review and editing. **Yong-Gang Chang:** Data curation; formal analysis; investigation; writing-review and editing. **Roger Tseng:** Data curation; formal analysis; investigation; writing-review and editing. **Sergey Ovchinnikov:** Formal analysis; investigation. **Rakefet Schwarz:**

Conceptualization; resources; writing-review and editing. **Andy LiWang:** Conceptualization; funding acquisition; project administration; supervision; writing-review and editing.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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